

MACHINE LEARNING PROJECT BRIEF

CUSTOMER CLICK PREDICTION

Predictive Modeling for Digital Ad Engagement

Problem Statement • Variable Description • Deliverables

Objective & Deliverables

Introduction

In the fast-paced world of digital advertising, marketing decisions increasingly depend on data-driven insight. This project focuses on predicting whether a customer will click on an online advertisement, using a robust machine learning model built on historical interaction data. The resulting model is intended to sharpen advertising strategy and improve the allocation of campaign resources.

Objective

Develop an effective machine learning model that predicts customer clicks on advertisements. By analyzing key behavioral and demographic features — including daily time spent on site, age, area income, and daily internet usage — the model will surface actionable insight into customer behavior and help improve the overall efficiency of ad campaigns.

Background

As the digital marketing landscape continues to evolve, understanding how customers engage with advertisements has become essential to campaign success. This initiative is motivated by the need to optimize ad targeting and spend, ensuring maximum impact and return on investment.

Expected Outcomes

The resulting model will function as a predictive tool that surfaces the factors most strongly associated with customer clicks. Beyond click prediction itself, the analysis is intended to support strategic decision-making and position the advertising function to run more targeted, effective campaigns going forward.

Recommended Project Steps & Guidelines

- **Data Exploration and Cleaning** Explore the dataset; identify and handle missing values, outliers, and duplicates. Build an understanding of the distribution of each feature.
- **Data Preprocessing** Encode categorical variables, scale numerical variables, and address any class imbalance. Split the data into training and test sets for model development.
- **Exploratory Data Analysis (EDA)** Conduct EDA to uncover patterns and relationships between user characteristics and ad-click behavior.
- **Feature Engineering** Create new features or transform existing ones to strengthen the model's predictive performance.
- **Build Classification Model** Develop a logistic regression model to estimate the probability that a given user clicks on an ad.
- **Model Evaluation** Assess model performance using accuracy, precision, recall, and F1-score. Iteratively refine model parameters to optimize predictive accuracy.
- **Reporting and Visualization** Produce a comprehensive report summarizing key findings, supported by visualizations that illustrate model performance and the insights gained.

Variable Description

The dataset provided for this project captures user interactions with online advertisements. Each row represents a unique user session; the table below describes each column.

Variable	Type	Description
Daily Time Spent on Site	Numeric	Time, in minutes, that a user spends on the website where the ad is displayed.
Age	Numeric	Age of the user who interacted with the advertisement.
Area Income	Numeric	Income level of the geographical area in which the user is located.
Daily Internet Usage	Numeric	Time, in minutes, that a user spends on the internet daily.
Ad Topic Line	Text	Topic or content of the advertisement the user interacted with.
City	Categorical	City of residence of the user.
Male	Binary	Gender indicator (1 = male, 0 = female).
Country	Categorical	Country in which the user is located.
Timestamp	DateTime	Date and time at which the user interacted with the advertisement.
Clicked on Ad	Binary	Target variable: whether the user clicked on the ad (1 = clicked, 0 = not clicked).

Note: "Clicked on Ad" is the target variable for the classification model; all other fields are candidate predictors or identifiers to be evaluated during feature engineering.